

Manasvi Agarwal

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Education

Thapar Institute of Engineering and Technology, B.E. Computer Engineering

- CGPA: 8.95 (Till 5th Semester)

JEE Mains

- Percentile: 95.8

Skills

Languages: C, C++, SQL, Python

Technologies: Git, CI/CD, AWS, Jenkins

Core Subjects: DBMS, Data Structures and Algorithms, OOPs, Operating Systems, Artificial Intelligence, Machine Learning, Software Engineering, Computer Networks, DevOps

Projects

Telecom RAG Assistant and DevOps Pipeline

- Built an end-to-end Retrieval-Augmented Generation (RAG) system capable of answering telecom-related queries from **50+ telecom documents**
- Processed and indexed **10,000+ text chunks** using Recursive Character Text Splitting, Sentence Transformers, and ChromaDB for semantic search
- Developed RESTful APIs using FastAPI for document ingestion, indexing, retrieval, and AI-powered question answering, achieving **sub-2-second response time**
- Containerized the application using Docker and automated testing/build workflows through GitHub Actions CI/CD pipelines
- Deployed the solution on AWS and integrated Prometheus-Grafana dashboards for real-time monitoring and observability

AI-Based Insurance Cost Prediction

- Built ML model to predict insurance cost using features like age, BMI, income and lifestyle
- Processed dataset of **1000+ records** with preprocessing, feature encoding, and training
- Improved prediction accuracy by **20%** using optimization techniques

DevOps CI/CD Monitoring Dashboard

- Built a web-based dashboard using the GitHub REST API to fetch and display live repository data
- Integrated **Jenkins CI/CD pipelines** to monitor **50+ builds** and deployment workflows
- Displayed commits, pull requests, issues, and pipeline/workflow status in real-time
- Reduced manual monitoring effort by **35%** through automation

Certifications

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| • AWS Academy Graduate – Cloud Foundations, Amazon Web Services | Jan 2026 |
| • AWS Academy Graduate – Microservices and CI/CD Pipeline Builder, Amazon Web Services | 2026 |

Honors and Achievements

- Winner – Agentic AI Hackathon (Innovation Award), organized by Ulster University, UK
- Selected among Top 100 teams in internal Smart India Hackathon round out of 500+ teams

Leadership and Activities

- Campus Ambassador – GirlScript Summer of Code (GSSoC)
- Convener – Literary Society (LITSOC), Thapar Institute of Engineering and Technology
- Anchor – Mudranight event, hosted audience of 500+ students